

Stock Price Prediction – ML Project Setup

Setup, Installation, Libraries, How to Run, Project Flow

Overview

This project predicts future stock prices using an **LSTM (Long Short-Term Memory)** deep-learning model.

It includes:

- Data extraction using **Yahoo Finance**
- Data preprocessing & scaling
- LSTM model training
- Prediction visualization
- A full **Streamlit web app** (app.py) for interactive use

Project Structure

|— app.py

|— keras_model.h5

|— Stock_Price_Prediction.ipynb

- **app.py** → Main Streamlit application
- **keras_model.h5** → Trained LSTM model
- **Stock_Price_Prediction.ipynb** → Notebook showing training code & analysis

Installation Guide

Install Python

Install **Python 3.8+**

<https://www.python.org/downloads/>

Create a Virtual Environment (recommended)

```
python -m venv venv
```

```
source venv/bin/activate    # On macOS / Linux
```

```
venv\Scripts\activate      # On Windows
```

Install Required Libraries

Run:

```
pip install numpy pandas matplotlib pandas_datareader yfinance keras tensorflow streamlit
```

If TensorFlow fails:

```
pip install tensorflow==2.12
```

Download the Project Files

Extract the zip and open the project folder.

Run the Application

Inside the project folder:

```
streamlit run app.py
```

This launches a local web UI in your browser where you can:

- Select a stock (ex: AAPL, TSLA, INFY)
- View historical price data
- See predictions from the trained LSTM model

Libraries Used

Library	Purpose
numpy	Numerical operations
pandas	Data processing
matplotlib	Plotting graphs
pandas_datareader	Fetching financial data
yfinance	Stock data download
keras / TensorFlow	Deep learning & LSTM model
streamlit	Web app UI
warnings	Clean console logs

How It Works

1. Downloads stock data (2015–2025)
2. Preprocesses & scales values
3. Splits data into train/test sets
4. Loads pre-trained LSTM model (keras_model.h5)
5. Predicts next-day prices
6. Reverses scaling
7. Visualizes:
 - Actual prices
 - Predicted prices